Hillcrest (Wine Glass): our busiest relay, and why it looked like our worst

Investigation 03 · 4 November 2026 · Hillcrest AL (Wine Glass) (540235, rain 6200, battery 6201), repeater on 151.500 MHz, Brisbane hub · Open: critical flag not borne out, a small fault remains

- **Hillcrest brings us most of 151.500.** 74% of the bursts we measured a carrier for over 13 days came on its carrier, near +170 Hz. The addresses only it relays come there; the ones only Constitution Hill relays don't.
- **It never went off air.** Every three-hour block had bursts on its carrier. From 24 October to 2 November our receiver heard the whole channel about 1.8 dB nearer the noise. Hillcrest reaches us weaker than most, so it lost the most: its gauge's three-hourly reports went from 93% heard to 41%, then back to 82%.
- **"70.8% corrupted" is a counting error.** It divides 80 corrupted copies by only the 113 bursts credited to Hillcrest itself. On its whole carrier the share is 2.9%, and 87% of those were one burst read two ways by our own decoder.
- **What's left is small but real.** 1.6% of the relays we could check came through damaged, against 0.2% on the channel's other carriers. That damage was done before the burst reached us.

Whose carrier it is

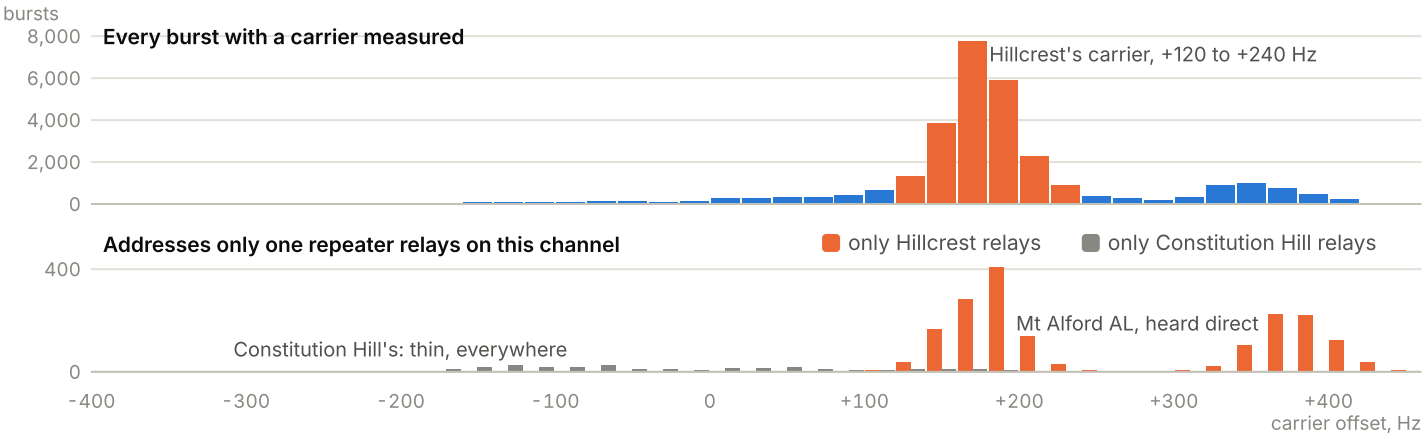


Figure 1. Bursts on 151.500 MHz by carrier offset, 22 October to 3 November, in 20 Hz steps (a small group near +2.5 kHz is left out). Below: only the addresses one repeater relays on this channel. Hover a bar for its counts.

74% of the bursts with a carrier measured, and 51% of all we decoded on the channel, came on one carrier between +120 and +240 Hz. Hillcrest's pass ranges overlap Constitution Hill's almost entirely, but two groups of addresses tell them apart. 23 of the 27 addresses only Hillcrest relays here came mostly on this carrier. The exceptions are Mt Alford AL, heard direct near +380 Hz, and two addresses heard fewer than ten times. Of 228 bursts from addresses only Constitution Hill relays, 34 (15%) fell there, about what any 120 Hz of their spread holds. 236 of 240 reports from Hillcrest's own gauge and battery came on it too. Flood-Net's carrier table missed this. It credits 14,942 of these bursts to 107 transmitters as if each were heard direct, and Hillcrest with only its own 110.

Never off air: our receiver heard the channel weaker

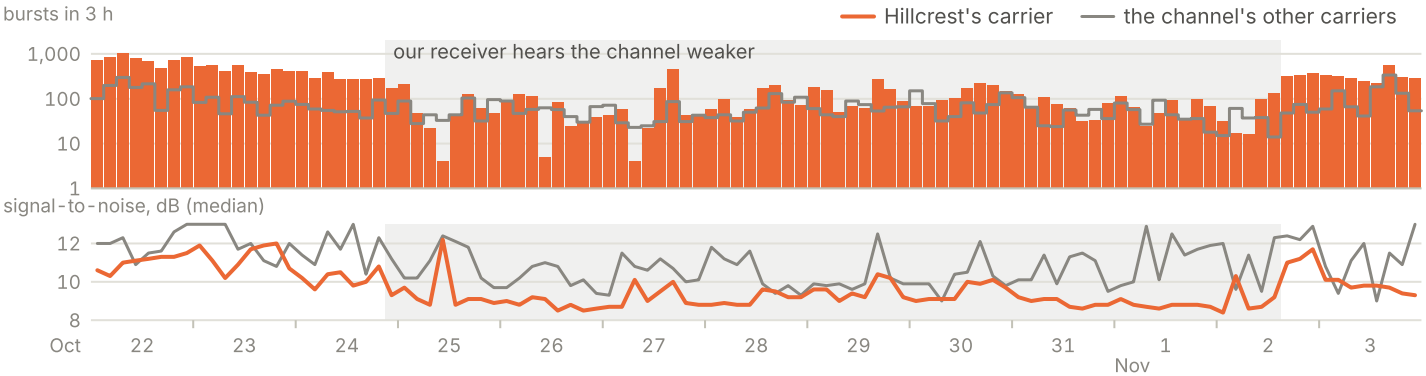


Figure 2. Top: bursts every three hours on Hillcrest's carrier (bars) and the channel's other carriers (line), log scale. Bottom: the median signal-to-noise of each. Shaded: the weaker spell.

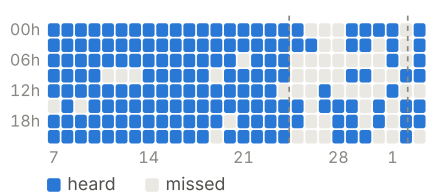


Figure 3. Hillcrest's three-hourly reports, 7 October to 3 November. Dashed: 21:00 on 24 October and 15:00 on 2 November.

The quietest three hours on its carrier still had 4 bursts. The "off air" Station Health saw was its gauge's longest silence, 22 hours from 14:15 on 26 October, while the carrier relayed 216 bursts.

From the evening of 24 October the median signal-to-noise fell from 10.9 to 9.1 dB on Hillcrest's carrier, and from 12.0 to 10.2 dB on the others. They fell together, so the cause is something they share. At our receiver the noise floor rose 3.7 dB, and burst levels only 2.3 dB. On the afternoon of 2 November levels rose 5.2 dB and the floor 4.7 dB. The channel came part of the way back (9.8 and 11.5 dB), and the gauge came back with it: the bulletin's "back on air". We heard 133 of 143 of its reports before, 29 of 70 during and 9 of 11 after.

Most of the corruption is our decoder

Our receiver stamps every frame it decodes from a burst with the moment that burst ended. 544 of the corrupted copies on Hillcrest's carrier share that moment, to the millisecond, with their clean twin. They were never sent: each is one burst read two ways. In 88% of these pairs the wrong frame had fewer votes than the right one (median 6 against 12). Our decoder drops a lookalike only when the real frame has three times its votes. The other carriers' pairs look the same (89%).

The weaker the signal, the more often it happens (Figure 4). At every strength from 8 to 15 dB it happens less on Hillcrest's carrier than on the others: 27 per 1,000 against 58 at 9 dB. Station Health called 79 of its 80 copies clear. But a copy from the same burst is always exactly as strong as its twin, so that test can't tell.

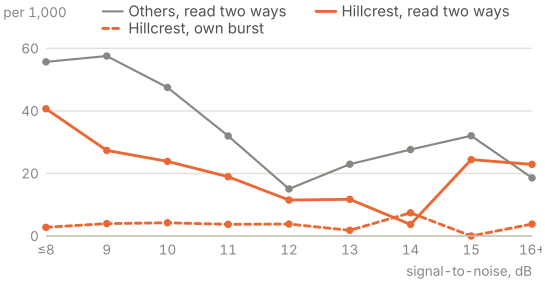


Figure 4. Corrupted copies per 1,000 decoded on 151.500, by signal-to-noise. Hover a point for its counts.

What is Hillcrest's

81 copies on its carrier were corrupted in a burst of their own, 0.3 to 5 seconds from a clean twin. The twin came on Hillcrest's carrier 36 times, on another 18 times and with no carrier measured 27 times. These hold at 2 to 4 per 1,000 from 8 to 13 dB; they don't rise as the signal weakens. The median copy was only 0.5 dB weaker than its twin. So the damage was done before the burst reached us. The copies are spread over 51 addresses, at most 7 from one station (Ipswich AL). 53 of the 56 that changed only the value read high, and 46 changed only its top bits. That is investigation 01's pattern: damage at the end of a frame, where no check bit sees it.

Counted	Corrupted	Out of	Share
Station Health's figure, 7 days to 10:13 on 4 November	80	113 credited to it	70.8%
Everything on its carrier, 22 Oct to 3 Nov	625	21,923	2.9%
The same, without bursts read two ways	81	21,923	0.4%
Relays on its carrier we could check	**81**	**4,977**	**1.6%**
The same, the channel's other carriers	5	2,169	0.2%
The same, 151.525 MHz (Toowoomba's channel, 02)	220	4,347	5.1%

How the flag shrinks. Station Health compares 70.8% with 1.27% for a typical transmitter, counted its way.

What we know

Established. The carrier near +170 Hz is Hillcrest's. It was on air throughout and carries most of what we hear on 151.500. The 70.8% comes from counting only 113 bursts. 87% of the corrupted copies on its carrier were never sent.

Likely. The quiet spell of 25 October to 2 November was at our end: Hillcrest's signal and the others' fell and recovered together, in step with our receiver's own readings. Hillcrest damages 1.6% of the relays we could check, about 7 times the rate on the channel's other carriers.

Open. What changed at our receiver on 24 October: a setting, or noise near it. Whether the 1.6% enters at Hillcrest's receiver or its transmitter. Why its battery reads 13.1 V at night and 13.2 V by day, with no solar charge showing.

Next

1. Fix Station Health's transmitter figures: count the whole carrier, and leave out bursts read two ways (or have the decoder drop a lookalike with fewer votes). The Bit-Flip Sources and Backbone tabs and the bulletins all read them (#272).
2. Teach the carrier table that +120 to +240 Hz on 151.500 is Hillcrest's. It now credits 107 transmitters there as heard direct (#271).
3. Find what raised our receiver's noise floor on 24 October, and keep the setting of 2 November (#273).
4. No urgent visit is needed. At the next one, check Hillcrest's power supply and its receive side.

Method: Flood-Net's receiver, legacy ALERT bursts on 151.500 MHz, 22 October to 3 November (Brisbane time), while carrier offsets were measured; Hillcrest's gauge from 7 October. A corrupted copy is one Flood-Net's checks found: a burst one or two bits from a real one, on the same receiver within five seconds. Hillcrest's carrier is +120 to +240 Hz. An address is Hillcrest's alone if it is in its pass ranges and in no other 151.500 repeater's in the station register. Data, queries and script: `data/investigations/2026-11-hillcrest-wine-glass/`, `tools/investigations/hillcrest_wine_glass.sql`, `tools/investigations/hillcrest_wine_glass.py`.

Words used here.

Carrier offset How far a transmitter's carrier sits from the channel's centre, as our receiver measures it. Each crystal is off by its own amount, so it tells transmitters on one channel apart.

Corrupted copy A second copy of a real report, heard within five seconds of it, with one or two of its address or value bits changed.

Own burst A corrupted copy that came in a transmission of its own, 0.3 to 5 seconds from its clean twin.

Votes In how many of our decoder's 45 guesses at a burst's carrier and timing a frame turned up. It reports a frame from 4.

Clear As strong as the clean copy, within 6 dB. Station Health takes it as sent wrong, not misheard.

Pass range The block of addresses a repeater is set to relay. Hillcrest passes 1900–2200, 4151–4199, 4800–4899, 5304–5399 and 6200–6274.

One burst read two ways A corrupted copy our decoder took from the same transmission as the clean one, at the same instant: one burst decoded twice, once wrongly.

Signal-to-noise How far a burst stood above the noise at our receiver, in decibels. Most bursts here arrive at 8 to 13 dB.

Checked The same reading reached us in another burst 0.3 to 5 seconds away, so a damaged copy shows.

Three-hourly report Hillcrest's gauge sends its count at 34 minutes past every third hour from midnight, rain or not, and its battery a few seconds later.